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Attribute Selection and Ensemble Classifier based Novel Approach to Intrusion Detection System

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Abstract

The rapid expansion of computer networks is causing vulnerabilities to occur, which in turn is compromising security. Thus, there is a need to monitor network traffic transmitted over these networks. Network vulnerabilities can cause huge loss to organizations. Hence, there is a need of effective and robust Intrusion Detection System (IDS) that can detect anomalies and raise alarm if such activity occurs. Recently, machine learning techniques have gained popularity for building intelligent Anomaly Detection Systems to detect novel attacks. These techniques produce diverse levels of accuracy and precision. Machine learning classifiers are first trained using training data to learn attack pattern of intrusions and then, tested against test data to differentiate normal data and anomalous data. The proposed work in this paper uses ranker-based attribute evaluation technique to reduce number of attributes and evaluates the implemented model using ensemble of IBk(K-NN), Random Tree, REP Tree, j48graft and Random Forest classifiers. The obtained experimental results show accuracy of 99.72% for binary classification and 99.68% for multi-class classification. The model uses NSL-KDD dataset for evaluating the performances. It is revealed by the experimental results that time complexity and computational expense are minimized due to dimensionality reduction of data and hybrid approach that is used for classification.

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Keywords: Computer Network Security, Intrusion Detection System, Machine Learning, NSL-KDD dataset.

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1. Introduction

Important data constantly attracts adversary and thus, is susceptible towards intense network invasion. When adversary forwards malicious packets to host system or to steal or change confidential data via a compromised network, then it is termed as intrusion. Intrusion Detection Systems (IDS) are commonly divided into two types: Signature and Anomaly IDS. Signature-based intrusion identification rely on comparison with signatures of recognized attacks which are stored in a database, but it cannot detect unknown attacks. However, Anomaly-based IDS use statistical approach to detects activities that deviates from usual limit of resource use and typical conduct parameters. False positives and False negatives rate remain high in case of anomaly-based identification. A successful intrusion consists of following steps [1]:

- Reconnaissance: Act to explore specially to gain information about the target. It is achieved by executing network commands such as "nslookup", "whois" to get domain name, IP addresses, server details etc.
- Scanning and probing: Includes finding insecure areas in target system to search for valuable information.
- Remote to local attacks (R2L): To gain access by remotely sending malicious network packets so that attacker can run commands on the target system. These attacks are generally performed by taking advantage of system vulnerabilities, using open ports, password prediction, sniffing etc.
- User to root attacks (U2R): These attacks are done for acquisition of administrator access of system for entire control.
- Start attacks: Attacks are instigated after getting root access. For example, stealing or modifying confidential information.

IDSs are deployed to raise an alarm if any such activity occurs in internal networks. Productive IDS are typically created through the usage of machine learning and data mining strategies, because these can incredibly distinguish intrusions. These procedures include a training phase in which model is trained using datasets. Datasets contains labelled samples of both attack and normal class. After training these models with mathematical algorithms, trained model is tested on separate samples of data to check accuracy of prediction.

This paper is divided in 7 sections. After introduction, discussion about related works is given in section 2 and performance evaluation metrics are explained in section 3. Classification algorithms and feature reduction methods are described in section 4 and section 5, respectively. The proposed model is discussed in section 6 of the paper, followed by analysis of results, in section 7. Finally, the paper is concluded in section 8.

2. Related Works

Dorothy E. Denning et al. implemented the first IDS in 1986 under research group named SRI International [2].. After that IDS became a mainstream issue of study in the research community.

In 2011, S.J. Horng, et al. [3] proposed an IDS model that used hierarchical clustering algorithm for attribute selection and Support Vector Machines (SVM) classifier for classification. The model was then tested on KDDcup 99 dataset. S. Mukherjee, et al. [4] in 2012 proposed an attribute subset evaluation method named Feature vitality-based reduction by using information gain, gain ratio and correlation-based feature selection. The work used Naïve Bayes classifier for testing of framework. R. M. Elbasiony [5] in 2013 came up with hybrid approach by combining Random Forest and weighted k-means classifiers and tested it on KDDcup99 data. In 2015 E. D. L. Hoz, et al. [6] used Principal Component Analysis (PCA) and Fisher Discriminant Ratio for attribute selection and noise reduction. Also, it used Probabilistic Self-Organizing Maps to build model and detect anomalies. Rajni Devi, et al. [7] in 2016 used K-NN and NN classifiers to give right answers to questions asked in Hindi language. The study was done on various datasets. Thaseen, I. S., et al. [8] in 2017 used chi-square attribute evaluator on NSL-KDD dataset and built a model that used Multi-class SVMs.

Jabbar, M. A., et al. [9] in 2017 proposed a model based on ensemble of Random Forest and Average one dependence estimator and used it against Kyoto dataset. H. Singh, et al. [10] in 2017 discussed tools which can be utilized by cyber security analysts to examine vulnerabilities. Belouch, M., et al. [11] in 2018 used UNSW-NB15 dataset to analyze multiple classifiers. H. Singh, et al. [12] in 2018 discussed existing cyber-attacks such as XSS, SQL injection and their impact on network security. Salo, F., et al. [13] in 2019 used Information Gain and Principle

Component Analysis for feature reduction on complex datasets and tested it against ensemble of Support Vector Machines, K-nearest neighbor and multi-layer perceptron classifiers.

3. Performance Measures

Following performance measures are used for evaluating performance of models.

- True-Positive (TP): Correctly classify an anomalous sample as attack.
- True-Negative (TN): Correctly classify a non-attack sample as ordinary instance.
- False-Positive (FP): Incorrectly classify an ordinary sample as anomalous instance.
- False-Negative (FN): Incorrectly classify an attack sample as ordinary instance.

Table 1. Confusion Matrix

		Predicted	
		Attack	Normal
Actual	Attack	TP	FN
	Normal	FP	TN

Using above mentioned terms and matrix from Table 1, several measures are used to determine performance and are examined underneath:

- Accuracy: It is the percentage of correctly predicted samples out of total samples in dataset.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (1)$$

- False Positive Rate: Rate of incorrectly classifying an ordinary sample as anomalous instance:

$$False\ Alarm\ Rate = \frac{FP}{FP+TN} \quad (2)$$

- Attack detection Rate: It is the number of samples that belong to positive class and are correctly classified as positive.

$$Attack\ Detection\ Rate = \frac{TP}{TP+FN} \quad (3)$$

- Precision: Precision is rate to find the likelihood that a positive prediction is correct.

$$Precision = \frac{TP}{TP+FP} \quad (4)$$

- ROC curve: ROC is utilized to draw curve between FPR and TPR. It is used to find cost sensitivity of the classifier.

4. Classification Algorithms

Classifiers are utilised to differentiate data between classes according to the research problem. Classifiers are of different categories and each category consists of various machine learning techniques. Some of these classifiers are briefly explained in this section.

4.1 Function Classifiers

These classifiers include neural networks and regression. These statistical models are based on biological neurons. Accuracy of these networks depends on the amount of training data. These use raw input data and maps it to output data. Example of these classifiers include RBF Network, SMO classifiers. Nonlinear functions are modelled easily by RBF classifiers. Raw input data is not used by this classifier. RBF Networks processes data iteratively similar to neural networks but RBF networks tends to over train [14].

4.2 Bayes Classifiers

In Bayes Classifiers, knowledge about uncertain domains is represented by the use of probabilistic graphical models. The nodes of graph denote random variables and probabilistic dependencies represents weights of edges which connect these random variable nodes. In these classifiers, class predicting depends on value of instances of attributes. Example of classifiers in this category are Naive Bayes, Bayes Net etc [14].

4.3 Lazy Classifiers

Lazy classifiers request to store complete training data for constructing classification model. Therefore, fresh data samples cannot be added during training time. These classifiers are robust and simple and effectively classify data streams [15]. Example of lazy classifier include K-NN (K Nearest Neighbour).

4.4 Meta Classifiers

In these classifiers, base classifier is first trained using best set of features and then it is used to make further predictions. Example of some meta classifiers are: LogistBoost, AdaBoost M1, Bagging, Rotation Forest, Vote, Grading, Classification via regression, Classification via Clustering etc.

4.5 Rules Classifiers

Rules Classifiers uses association rules for classification between various features. True classification is termed as coverage and quantified in terms of precision. Association rules used in classification are mutually exclusive [15]. Rules classifier can predict one or more results. Examples are: Zero R, OneR, JRip, Decision Table etc.

4.6 Trees

Also known as decision trees, these classifiers create a predictive and descriptive model that divide actual classification problem into sub-problems by creating a flowchart like tree model. Nodes in the tree represents input instance value and tree leaves denote the prediction of that input. Example of classifiers in this category are j48, Random Tree, j48 graft, Random Forest, Decision Stump, REP Tree etc.

5. Feature Reduction Methods

Majority of Machine learning and data mining approaches couldn't work well with intrusion detection because of gigantic complexity and size of datasets. These techniques take huge computational time to classify attacks which makes implementation more difficult in real time environments. This is because of huge number of features are contained in network data which is to be processed by Intrusion Detection System. For better classification, quantity and quality of features matter and it helps us understand their importance and their co-relation. If features selected are very less, then classification quality will reduce and if they are more than required, it will make loss of generalization. Experimental results show that accuracy and computational cost is improved when we use feature extraction techniques in Intrusion Detection Systems [16]. Thus, dimensionality and feature reduction strategies are being used as a pre-processing step to improve accuracy and to reduce time for attack detection.

5.1 Feature selection

Feature selection methods are utilised to discover smaller set of attributes which could improve overall outcome of the procedure and generate few errors. Another goal is to decrease computation time and storage utilization. In IDS, feature selection techniques are utilized to improve accuracy of attack detection. Some of the dominant feature selection methods are Principal component analysis (PCA), Information gain (IG), and genetic algorithm (GA) [17]. Filter and Wrapper are two kind of features selection strategies in which we incorporate different FS methods.

- Filter methods don't utilize any classifier for feature evaluation and are relatively powerful against overfitting, yet it utilizes autonomous estimation techniques, for example, distance measures, consistency measures, and correlation measure.
- In Wrapper technique, a classifier is employed as a black box for evaluating optimal features. These methods achieve great speculation, yet sometimes endure high dimensionality due to the computational expense of preparing the classifier.

5.2 Feature Extraction

In a dataset, rows represent samples and columns represents features, where features are a result of quantitative and subjective findings. Feature extraction is used to decrease the dimensionality of dataset by reducing set of attributes in a way that accuracy of attack detection is not altered and time used in discovery is reduced. Numerous feature extraction methods are available in the field. For example, self-organizing maps, principal component analysis etc.

5.3 Clustering

In clustering data samples are grouped into sets of data where data samples in each set is similar in one way or the other.

6. Dataset Description

NSL-KDD dataset is an improved form of KDDcup99 dataset [1]. NSL-KDD is proportionate to 10% of KDDcup99 dataset as it is comprised of huge amount of information. The instances in the dataset are categorized as attack or normal. Various machine learning models have been made and analysed using NSL-KDD dataset and discussed in [18]. To improve the data handling capabilities of IDS and deal with large size of NSL-KDD dataset, multicore hardware platforms appears to be appealing. Hardware acceleration is used in CNNs for application in image processing in [19]. Major modifications are performed on KDDcup dataset:

- Redundant records were removed so that classifier doesn't show impartial results.
- Records present are rational and sufficient amount of records are present in training and testing datasets.

Every sample in NSL-KDD dataset consists of 41 features or attributes. Classifying attribute is named either attack or normal. Attribute description such as name, data type, details are described in [18]. NSL-KDD dataset contains 39 different attack types which are divided in 4 attack classes i.e. DOS, Probe, U2R and R2L.

Out of 39 different attack types, there are 10 types of attacks in Denial of service (DoS) class, 6 type of attacks in Probe class, 16 types of attacks in R2L class and 7 types of attack in U2R. An overview of data samples of NSL-KDD dataset which are utilised in testing and training phase of developed IDS model are given in Table 2.

Table 2. NSL-KDD Dataset Overview

DATA SET TYPE	Number of Instances					
	Total Instances	Normal Instances	DoS Instances	Probe Instances	U2R Instances	R2L Instances
Training set	125973	67343	45927	11656	52	995
	%	53.46	36.45	9.25	0.04	0.79
Test set	22543	9711	7458	2421	200	2754
	%	43.08	33.08	10.74	0.89	12.22

7. Proposed Model

The proposed model contains following steps:

- For data capturing and feature selection, NSL-KDDTrain+20% is used which contains 25192 instances. Out of total instances, 13449 are normal and 11743 are attack instances.
- Data from NSL-KDD dataset is clustered into 4 datasets for multi-class classification i.e. classification of different type of attacks. For the experiment, the dataset is divided into 2 parts i.e. 80% for training purpose and 20% for testing purpose.
- Pre-processing and normalization of data is done before training. This phase is used for removal of noise or redundant information from data and preserve only meaningful and important information only [22]. In the proposed model, pre-processing involves the following main tasks:
    - Transformation of non-numerical attributes into numeric attributes using NumericTransform filter: In generic dataset attribute number 2 (protocol type), 3 (service), and 4 (flag) have non-numerical values. These attributes are converted into Numerical values by assigning a specific value to each variable (e.g. TCP = 1, UDP = 2 and ICMP = 3). This is done because decision tree classifiers can predict using numerical values only.
    - Transformation of anomaly into its type at the end of each instance is done by assigning numeric value to each attack type i.e. 1 to normal data, 2 to DoS attack, 3 to Probe attack, 4 to R2L attack and 5 to U2R attack respectively.
    - Normalization of data was done to make data values comparable. Thus, minmax normalization was used to ensure all attribute values lie in between [0,1].
- Information gain based attribute evaluation is used with ranker search method to reduce features from 41 attributes to just 11. Attributes over information gain of 0.40 were selected, i.e. 2,40,3,41,27,26,30,31,32,36 and 9.
- Hybrid model is build using ensemble of IBk, RandomTree, REPTree, j48graft and Random Forest classifiers and then training data is provided for training purpose.
- After training, model is provided with test data for evaluation to find performance measures such as accuracy, false positive rate, true positive rate etc.

Framework of proposed model is presented in Figure 1.

8. Experimentation and Results

8.1 Experiment setup

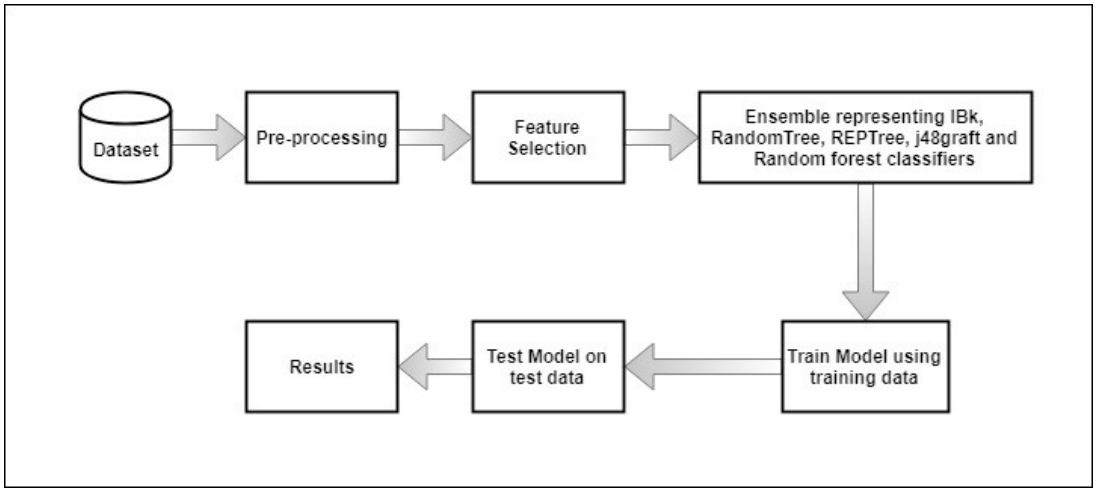


Figure 1. Proposed Model

Evaluation of the model is done using WEKA Knowledge analysis tool Version 3.8.3. WEKA is GUI based software suite written in java [17]. It has 49 tools for data pre-processing, 15 tools are for attribute evaluation, contains 76 classifiers for classification tasks and 10 feature selection methods. Weka supports multiple formats such as CSV (Comma Separated Values), ARFF (Attribute Relation File Format). Weka also provides tools for visualization of data. The proposed work uses 20% of total NSL-KDD dataset in this experiment and conducted binary and multi-class classification. Ranker-based search method and information gain is used for attribute subset evaluation. In the end, model is built using selected feature subset and ensemble of classifiers. Weka configuration presented below is used to build the proposed model:

weka.classifiers.meta.Vote -S 1 -B "weka.classifiers.lazy.IBk -K 1 -W 0 -A "weka.core.neighboursearch.LinearNNSearch -A \\\"weka.core.EuclideanDistance -R first-last\\\"\"" -B "weka.classifiers.trees.J48graft -C 0.25 -M 2" -B weka.classifiers.trees.RandomForest -P 100 -I 100 -num-slots 1 -K 0 -M 1.0 -V 0.001 -S 1" -B "weka.classifiers.trees.RandomTree -K 0 -M 1.0 -V 0.001 -S 1" -B "weka.classifiers.trees.REPTree -M 2 -V 0.001 -N 3 -S 1 -L -1 -I 0.0" -R AVG

8.2 Result Analysis

Outcomes of experimentation revealed that the proposed model exhibited better accuracy than conventional classification techniques and much better false positive rate. Proposed model gave accuracy of 99.72% for binary class classification and 99.68% for multi class classification. False positive rate of model was recorded as 0.003 and recall was measured as 0.997 which are best amid other classifiers. A comparison of performance measures is given in Table 3. Classifications of each attack type are compared in Table 4. A comparison of research work on NSL-KDD Dataset is presented in Table 5 for validation of proposed model. All classifiers were tested on reduced feature set.

Table 3. Comparision with traditional classifiers

Classification Algorithm	Accuracy	FP Rate	Recall
SVM	96.42%	0.036	0.964
Naïve Bayes	87.87%	0.127	0.087
IBk	98.07%	0.019	0.981
Proposed Model	99.72%	0.003	0.997

Table 4. Multi-class classification comparison

Classifier	Attack Class	Outcome
SVM	DoS	98.32
	Probe	98.31
	U2R	99.96
	R2L	96.08
Naïve Bayes	DoS	95.98
	Probe	91.83
	U2R	96.39
	R2L	98.38
IBk (K-NN)	DoS	99.88
	Probe	99.68
	U2R	99.88
	R2L	99.45
Proposed	DoS	99.99
Model	Probe	99.74
	U2R	99.92
	R2L	99.89

Table 5. Comparison with Related Research Work

Author(s)	Data Reduction Method Used	Classifier Used	Outcome/Accuracy
S. Mukherjee, et al. [4]	Feature Vitality Based Reduction Method	Naïve Bayes	97.78%
M. S. Pervez, et al. [20]	Self-proposed	SVM	99% (using 36 features)
R Singh, et al. [21]	Alpha Profiling	Online Sequential Extreme Learning Machine	97.67%
Thaseen, I. S., et al. [8]	Chi-square	SVM (Multiclass)	98%
Salo, F., et al. [13]	Information Gain and Principal Component Analysis	Ensemble (SVM, IBK and MLP)	98.24%
Proposed Model	Information Gain	Ensemble (IBk, RandomTree, REPTree, j48graft and Random Forest)	99.72%

Visualisation of ROC Curve for Normal class in case of binary classification is shown in Figure 2. In the ROC Curve below, it can be seen that False positive rate is represented on x-axis and True positive rate is represented on Y- axis. Area under ROC Curve is 0.999 which clearly indicates that proposed model is suitable to classify the anomalies present in dataset.

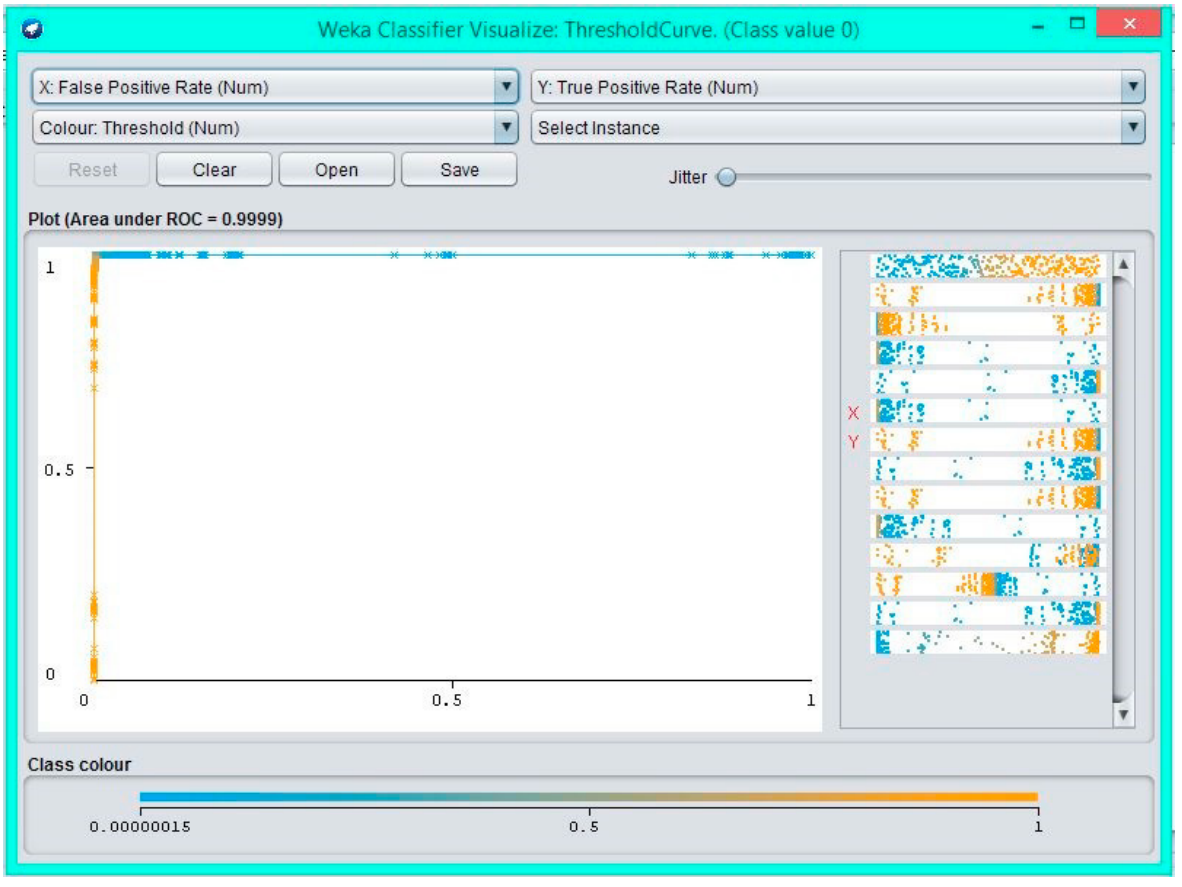


Figure 2. Visualisation of ROC Curve for Normal Class

9. Conclusion and Future Scope

In present scenario, intrusion detection models are critical or important for network security and using machine learning techniques for the same have given a major boost to research in this area. Application of ensemble learning methods for increasing accuracy and robustness is the aided advantage. The work proposed in this paper used filter-based attribute evaluation technique along with ensemble classifier and found that the proposed model is robust enough to detect intrusions with minimal computation expense. Model accuracy has been improved and false positive rate has been found negligible. In future, the work can be extended by using wrapper-based attribute evaluation for optimizing the model and simulating the same in real-time environment to check its actual effectiveness.

References

[1] Shadi Aljawarneh, Monther Aldwairi, and Muneer Bani Yassein. (2018). "Anomaly-based intrusion detection system through feature selection analysis and building hybrid efficient model." Journal of Computational Science, **25**, 152-160.

[2] Dorothy Elizabeth Denning, "An Intrusion-Detection Model," in IEEE Transactions on Software Engineering, **vol. SE-13, no. 2**, pp. 222-232, Feb. 1987.

[3] Shi-Jinn Horng, Ming-Yang Su, Yuan-Hsin Chen, Tzong-Wann Kao, Rong-Jian Chen, Jui-Lin Lai, and Citra Dwi Perkasa.. (2011). "A novel intrusion detection system based on hierarchical clustering and support vector machines." Expert systems with Applications, **38(1)**, 306-313.

[4] Saurabh Mukherjee and Neelam Sharma. (2012). "Intrusion detection using naive Bayes classifier with feature reduction." Procedia Technology, **4**, 119-128.

[5] Reda M. Elbasiony, Elsayed A. Sallam, Tarek E. Eltobely and Mahmoud M. Fahmy. (2013). "A hybrid network intrusion detection framework based on random forests and weighted k-means." Ain Shams Engineering Journal, **4(4)**, 753-762.

[6] Eduardo De la Hoz, Emiro De La Hoz, Andrés Ortiz, Julio Ortega, Beatriz Prieto. (2015). "PCA filtering and probabilistic SOM for network intrusion detection." Neurocomputing, **164**, 71-81.

[7] Rajni Devi and Mohit Dua. (2016) "Performance evaluation of different similarity functions and classification methods using web based hindi language question answering system." Procedia Computer Science, **92**, 520-525.

[8] Ikram Sumaiya Thaseen and Cherukuri Aswani Kumar. (2017). "Intrusion detection model using fusion of chi-square feature selection and multi class SVM." Journal of King Saud University-Computer and Information Sciences, **29(4)**, 462-472.

[9] M. A. Jabbar and Rajanikanth Aluvalu. (2017). "RFAODE: A novel ensemble intrusion detection system." Procedia computer science, **115**, 226-234.

[10] Himanshi Singh and Mohit Dua. (2017) "Detection & prevention of website vulnerabilities: Current scenario and future trends", 2017 2nd International Conference on Communication and Electronics Systems (ICCES), IEEE.

[11] Mustapha Belouch, Salah El Hadaj and Mohamed Idhammad. (2018). "Performance evaluation of intrusion detection based on machine learning using Apache Spark." Procedia Computer Science, **127**, 1-6.

[12] Himanshi Singh and Mohit Dua. (2018) "Website Attacks: Challenges and Preventive Methodologies", 2018 International Conference on Inventive Research in Computing Applications (ICIRCA), IEEE.

[13] Fadi Salo, Ali Bou Nassif and Aleksander Essex. (2019). "Dimensionality reduction with IG-PCA and ensemble classifier for network intrusion detection." Computer Networks, **148**, 164-175.

[14] Ranjita Kumari Dash. (2013). "Selection of the best classifier from different datasets using WEKA." International Journal of Engineering Research and Technology, **2(3)**.

[15] Tanya Garg, and Surinder Singh Khurana. (2014, May). "Comparison of classification techniques for intrusion detection dataset using WEKA." In International Conference on Recent Advances and Innovations in Engineering (ICRAIE-2014) (pp. 1-5). IEEE.

[16] Fatemeh Amiri, MohammadMahdi Rezaei Yousefi, Caro Lucas, Azadeh Shakery and Nasser Yazdani. (2011). "Mutual information-based feature selection for intrusion detection systems." Journal of Network and Computer Applications, **34(4)**, 1184-1199.

[17] Jorge R. Vergara and Pablo A. Estévez. (2014). "A review of feature selection methods based on mutual information." Neural computing and applications, **24(1)**, 175-186.

[18] L. Dhanabal and S. P. Shantharajah. (2015). "A study on NSL-KDD dataset for intrusion detection system based on classification algorithms." International Journal of Advanced Research in Computer and Communication Engineering, **4(6)**, 446-452.

[19] Dan Cireşan, Ueli Meier and Jürgen Schmidhuber. (2012). "Multi-column deep neural networks for image classification." arXiv preprint arXiv:1202.2745.

[20] Muhammad Shakil Pervez and Dewan Md Farid. (2014, December). "Feature selection and intrusion classification in NSL-KDD cup 99 dataset employing SVMs." In The 8th International Conference on Software, Knowledge, Information Management and Applications (SKIMA 2014) (pp. 1-6). IEEE.

[21] Raman Singh, Harish Kumar and R. K. Singla. (2015). "An intrusion detection system using network traffic profiling and online sequential extreme learning machine." Expert Systems with Applications, **42(22)**, 8609-8624.

[22] Frans Botes, Louise Leenen and Retha De La Harpe (2017). "Ant Colony Induced Decision Trees for Intrusion Detection." 16th European Conference on Cyber Warfare and Security, pp.53-62.